



A Comprehensive Guide on NuSTAR Data Processing and Analysis

Workshop on Astrophysics and Cosmology 2026

SXPC-NEPAL

Subhash Bhandari¹

Department of Physics
St. Xavier's College, Maitighar, Kathmandu, Nepal

¹cosubhasmos@gmail.com

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1 Introduction

This guide provides a comprehensive step-by-step tutorial for setting up and using HEASOFT, XSPEC, and related tools for astrophysical data analysis, focusing on black hole low-mass X-ray binaries (BH LMXBs), incorporating data provided by NuSTAR. It covers data downloading, software installation, calibration, and basic analysis techniques as outlined in the workshop curriculum. Before jumping into the software part, we will try to understand some basics about what we are actually doing and how it works. But if you want to dive right into the installation part, please refer to Section 5.

1.1 What Are BH LMXBs?

Black Hole Low-Mass X-ray Binaries (BH LMXBs) are fascinating astronomical systems consisting of a black hole (BH) and a low-mass companion star, typically with a mass less than or equal to about 1 solar mass ($M_{\odot}$). In these binaries, the black hole accretes material from the companion star via Roche lobe overflow or stellar winds, forming an accretion disk around the BH. This process heats the infalling matter to extreme temperatures, resulting in intense X-ray emissions, especially during outburst phases. These systems are crucial for understanding black hole formation, evolution, and spin properties [4]. Galactic BH LMXBs are believed to form through the evolution of isolated binaries or even triple systems, providing insights into the elusive black hole population in the Milky Way. Observations of their X-ray spectra help probe accretion physics, neutron star vs. black hole distinctions, and potential natal kicks during BH formation.

1.2 Importance in Astrophysical Research

Studying BH LMXBs allows researchers to:

- Understand the characteristics of the system, like spin, mass, and accretion properties of a black hole.
- Investigate outburst mechanisms and X-ray emission from accretion disks.
- Contribute to population studies of galactic black holes, revealing formation pathways like binary evolution or triple interactions.
- Link to broader cosmology by understanding compact object mergers detectable via gravitational waves.

In this workshop, we emphasize hands-on analysis using real data from missions like NuSTAR, which provides high-energy X-ray observations ideal for probing these systems.

1.3 About NuSTAR

NuSTAR is NASA's space-based X-ray observatory specializing in high-energy (hard) X-rays, making it ideal for probing extreme astrophysical environments like BH LMXBs during outbursts [2]. This mission was launched on June 13, 2012, via a Pegasus XL rocket from Kwajalein Atoll. It was NASA's 11th Small Explorer (SMEX) mission. Figure 1 shows its neat structure. It consists of two co-aligned grazing-incidence telescopes with cadmium zinc telluride (CZT) detectors. It observes in the broadband energy range 3–79 keV (focused), and extendable to 2.5–100 keV with a field of view: $12' \times 12'$, angular resolution 58" HPD (half-power diameter). The main objective of this telescope is to study

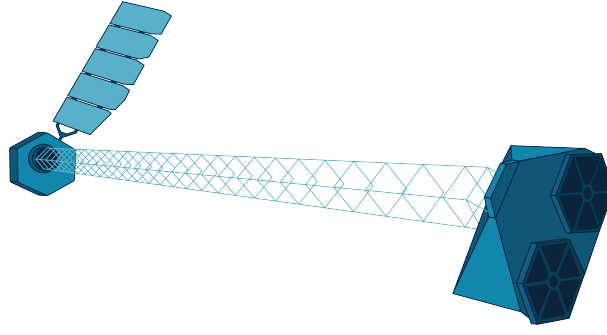


Figure 1: *An illustration of NASA's NuSTAR space telescope. The colors depicted in this illustration are for artistic purposes only. Credit: NASA*

black holes (formation, spin, accretion), neutron stars, supernova remnants, active galactic nuclei (AGN), and transient events like gamma-ray bursts or X-ray binaries. It resolves the cosmic X-ray background and detects faint, hard X-ray sources. NuSTAR excels at hard X-ray spectroscopy of BH LMXB outbursts, capturing Comptonized emission from coronae and reflection features (e.g., Fe K lines) to measure BH spins and disk geometry.

1.4 NuSTARDAS

NuSTARDAS or NuSTAR Data Analysis Software is a specially developed software to handle the full analysis workflow from raw telemetry data files (also called Level 0) to science-ready products (Level 2/3), including cleaning, calibration, and product generation. The software is designed as a collection of software modules (tasks), each dedicated to a single function. The package also includes a main script, the NuSTAR pipeline² (named `nupipeline`), allowing users to automatically run in sequence all the tasks for the data processing. Apart from that, it also contains `nuproducts` to generate spectra, light curves, and images, `nuexpomap` to create exposure maps, and others like `numkarf` (ARFs), `nulccorr` (deadtime correction), `nucalcsaa` (SAA filtering). It relies on CALDB for PSF, effective area, and clock corrections.

The NuSTAR data analysis is carried out in three primary stages:

- Data Calibration: FITS formatted telemetry processing to produce calibrated event files.
- Data Screening: calibrated event files filtering by applying cleaning criteria on specified orbital/instrument parameters and event properties to produce cleaned event files.
- Products Extraction: generation of high-level scientific products (light-curves, energy spectra, sky images, Ancillary Response Files, Redistribution Matrix Files).

The above steps will give us scientific data files often categorized into different levels as shown in Table 1. NuSTARDAS is integrated with HEASOFT as a subpackage within it. Therefore, its installation is also a part of the HEASOFT download. We will learn more about HEASOFT in the next section. In conclusion, the key task of this software is to do screening for bad pixels/SAA³ passages, perform barycenter/timing corrections ($<100 \mu\text{s}$ accuracy), and handle bright sources ($>100 \text{ cps}$) with custom filters.

²we will use `nupipeline` as a single task instead of multiple individual tasks

³South Atlantic Anomaly

File Type	What do they mean?
Level 1	Telemetry formatted in FITS format
Level 1a	Calibrated Event Files
Level 2	Cleaned and Calibrated Event Files
Level 3	High-level Scientific Products (light-curves, energy spectra, etc.)

Table 1: levels of science data

1.5 SAO Image DS9

SAOImage DS9 is an application designed for astronomical imaging and data visualization [3]. We will use DS9 to visualize astronomical sources and to extract background and source regions for cleaning NuSTAR data. DS9 supports FITS images and binary tables, allows for multiple frame buffers, enables region manipulation, and offers various scaling algorithms and colormaps. We will utilize event files to extract the regions using DS9.

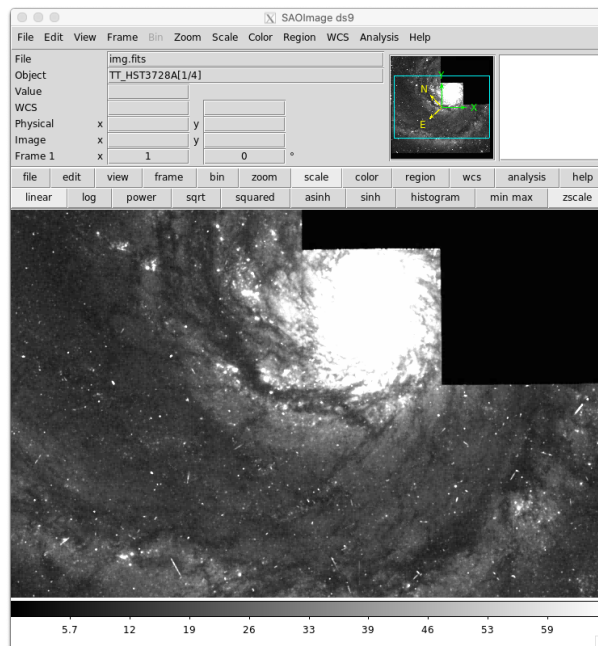


Figure 2: Snapshot of DS9 Software. Image: SAOImageDS9

1.6 HEASoft

HEASoft, High Energy Astrophysics Software, is our major sword in this analysis. We will perform everything with this software, and it is one of the best software in astrophysical data processing. HEASOFT operates within the HEADAS build environment and consists of two main components: a set of mission-independent utilities (such as HEATOOLS and HEASPTOOLS) and multi-mission utilities (like HEAGEN and ATTITUDE) developed and maintained by the HEASARC, as well as a collection of mission-specific packages managed, at least in part, by external groups. The core HEASARC utilities are typically written in ANSI C to ensure maximum portability. However, missions are free to use any programming language they choose for their own packages, keeping in mind that some language choices may restrict the platforms and/or compilers on which their tasks can be built and executed. HEASoft supports multiple missions, NuSTAR, Chandra, XRISM, Swift, Fermi, IXPE, TESS, etc. So if you want to work with any of that data, you will

need to have HEASoft installed a priori. We will discuss the installation and usage of HEASoft in a later section.

1.7 CALDB

CALDB, or Calibration Database, provides calibration files essential for accurate data reduction. Managed by HEASARC, it stores and indexes datasets related to the calibration of high-energy astronomical instruments. Users and software can easily access CALDB to find available calibration datasets and identify which ones should be used for data reduction and analysis. CALDB is versatile and compatible with various computer system environments, allowing for the easy addition of new missions and instruments. It can be installed and used at any location. The NuSTAR software tasks retrieve the calibration files structured in HEASARC's calibration database (CALDB) system. In our context, we will set up CALDB for NuSTAR only. You can add to other missions once you learn for any one of them.

1.8 XSPEC

XSPEC is a command-line X-ray spectral fitting package used for fitting theoretical models to X-ray spectra, which is essential for extracting physical parameters from black hole low-mass X-ray binaries data [1]. XSPEC is detector-independent and is used for missions like NuSTAR, Chandra, and XRISM. It has 200-plus built-in models, including additive models like powerlaw, diskbb, multiplicative models like phabs absorption, and convolution models like relconv for relativity. It has a Python version called PyXspec in which a user can run XSPEC with Python language scripts or interactively at a Python shell prompt. In this document, we will focus on XSPEC only. Once you finish extracting scientific products using HEASoft, it's time to fit and get something scientific from it. And that's where XSPEC comes in. XSPEC comes with HEASoft installation; we don't have to separately install it.

1.9 Workshop Context and Guide Objectives

This guide complements the workshop's astrophysics module, where we delve into the theory and practical data handling of BH LMXBs. Our goal is to equip you with the skills to download, process, and analyze X-ray data independently. By the end, you'll be able to perform spectral fitting and statistical analysis, fostering research in accretion physics and compact objects.

The guide is structured to mirror the data analysis workflow:

- Prerequisites and setup.
- Data acquisition.
- Software installation and configuration, including HEASoft, CALDB, XSPEC.
- Data calibration and preparation using HEASoft.
- Modeling and fitting in XSPEC.
- Advanced analysis and visualization.

2 Prerequisites

To follow this guide effectively, ensure you have:

- A Unix-like operating system (Ubuntu preferred; Windows users can use WSL2 or VirtualBox for a Linux environment).
- Basic familiarity with command-line interfaces (e.g., bash).
- Internet access for downloading software and data.
- At least 10-15 GB of free disk space for software, CALDB, and datasets.
- Installed dependencies: gcc, g++, make, perl, libncurses. We will install them here too.

Are you a Windows user?

The current^a version of HEASOFT doesn't run on a Windows environment. Even if they run, they don't run with full functionality. So, it is recommended to install VirtualBox, or double-boot Linux alongside Windows, or use Windows Subsystem for Linux (WSL). All of these options are hard and usually tricky to get software working right. To be honest, it would be better for an astrophysicist like you to switch to Linux. I have a pendrive ready. Just saying!

^aOlder ones too

3 Selecting Source

Selecting a source is a crucial step in this analysis. You should choose a source that you are interested in working with or one that has recently undergone an outburst. In this tutorial, we will use the source MAXI J1820+070 as an example. MAXI J1820+070 is a Black Hole Binary (BHB) system that consists of a black hole with an estimated mass of 7 to 8 solar masses, along with a companion star that is less than one solar mass [5]. Its distance from the Solar System is approximately 3.5 kiloparsecs, or about 10,000 light-years [6].

In such a BHB system, matter is transferred from the companion star to the black hole through an accretion disk. Understanding this process is essential for studying accretion physics and other characteristics, such as spin and mass. This is the main objective of our workshop. We will examine the accretion properties and behavior of the system to gain insights into the BHB dynamics, using spectral analysis of the source derived from specific data and a straightforward⁴ software tool.

4 Downloading Astrophysical Event Datasets

Event datasets for BH LMXBs can be sourced from archives like NASA's HEASARC or NuSTAR data repository. We will download the data for the source MAXI J1820+070 from the HEASARC archive. HEASARC is a NASA's designated multi-mission astronomy archive for cosmic astronomy datasets for observations in the extreme ultraviolet (EUV),

⁴but sometimes strenuous

X-ray, and gamma-ray wave bands, as well as for cosmic microwave background (CMB) datasets. Here's a quick steps to download the data from the archive:

1. Visit the HEASARC Browse interface: <https://heasarc.gsfc.nasa.gov/cgi-bin/W3Browse/w3browse.pl>.
2. Type object name MAXI J1820+070 in Object Name or Coordinates.
3. Select on NuSTAR on the missions and catalogs section.
4. Go all the way down and click on start search.
5. Now you will see all the available tables and row counts. Click on table name *numaster: NuSTAR Master Catalog* to view search results.
6. Select on the first row, with an observation id 90401309039.
7. Scroll to the bottom and retrieve data products for selected rows.
8. You will now see download options. Click on *Create Download Script* to download from the terminal or directly from the browser by clicking on *Retrieve*. Make sure you download it in your desired working directory

Tip

You may encounter a download error while downloading from the website if the expected file exceeds the limit of 2 GB per tar file. There are two options: either to go back and reduce the number of observations or files selected, or select Create Download Script to create an editable script of download commands. It would be the right option to create a download script and run it in the terminal. But in our case, the data is < 2 GB, so we will prefer downloading directly from the browser.

5 Setting Up Software

This section covers installing HEASOFT, NuSTARDAS, XSPEC, and CALDB. Users are recommended to follow the steps precisely as instructed. During installation, I will assume you have a working Linux environment, either on VirtualBox or WSL.

5.1 HEASOFT

5.1.1 Downloading HEASOFT

We can download HEASoft in different ways, either by downloading the source code for the software and compiling the tools or downloading precompiled executables for the most widely used computer platforms. We will download the latest version of HEASoft 6.36 in this workshop. Follow these steps to download HEASoft:

1. Visit <https://heasarc.gsfc.nasa.gov/docs/software/lheasoft/download.html>
2. In Step 1, click on "source code" and select your type of operating system (e.g., PC-Linux-Ubuntu).
3. In Step 2, you have to select the missions for which you are using this software for. In our case, we will use NuSTAR, so select NuSTAR along with XSPEC. Other required tools will automatically get selected once you select the mission type.

4. Click on submit, and the download process will automatically start.

Alternatively, you can use `wget` or `curl` if you are familiar with them, but we will not discuss them here⁵.

5.1.2 Prerequisites

At the very beginning, copy the downloaded folder(it's probably `heasoft-6.36src` or something like that) into the directory where you want to work on. Then unzip the downloaded folder.

```
tar xzf heasoft-6.36src.tar.gz
```

Make sure the filename is correct. It will take a minute or some to unzip the folder, which depends on your system's performance. Once it is completed, you will now have to install some dependencies. It is better to double-check even if you have them already.

```
sudo apt-get -y install libreadline-dev libncurses5-dev ncurses-dev curl \
libcurl4 libcurl4-gnutls-dev xorg-dev make gcc g++ gfortran perl-modules-5.38 \
libdevel-checklib-perl libfile-which-perl python3-dev python3-pip \
python3-setuptools python3-astropy python3-numpy python3-scipy python3-matplotlib
```

Now, there are a few Python environments which are required to use some tools and interfaces in HEASoft, such as HEASoftPy, PyXspec, HEASPy, IXPE, and NICER. Since we use none of them in this workshop, we will forgo installing them. Now it is time to ensure the correct compilers to make it smooth. Configure them by running these commands one by one:

```
export CC=/usr/bin/gcc
export CXX=/usr/bin/g++
export FC=/usr/bin/gfortran
export PERL=/usr/bin/perl
export PYTHON=/usr/bin/python3
```

External packages (e.g., Miniconda) may set compiler or other flags in the environment, which can negatively impact a HEASoft build, so you are advised to unset all `*FLAGS` and `conda` alias variables. Run the commands below to do so.

```
unset CFLAGS CXXFLAGS FFLAGS LDFLAGS build_alias host_alias
export PATH="/usr/bin:$PATH"
```

5.1.3 Configure

With the above steps completed and the appropriate environment variables set, now it's time to configure the software. Run the command below each at a time

```
cd heasoft-6.36/BUILD_DIR/
./configure > config.txt 2>&1
```

The configuration process may take several minutes to complete, depending on the performance of your device. After the run stops, run the command

⁵I am just too lazy

```
less config.txt
```

and check the last line of the file. If it is successful, the last line of output (in config.txt) should read "Finished"⁶. If you are too lazy to scroll all the way down, enter the shortcut **SHIFT+F** to skip to the bottom of the file.

5.1.4 Build

If and only if the configuration was successful, proceed to build the software. Run the following command in the terminal.

```
make > build.txt 2>&1
```

This process may take an hour or more to run, depending on your hardware and HEASoft download selections. If it is successful, the last line of output (in build.txt) should read "Finished make all". Check it again by

```
less build.txt
```

Again, if it is not successful, do not proceed until any errors have been resolved.

5.1.5 Install

If only the previous step, 'make', is successful, you can proceed to install the software. Run the following command to install:

```
make install > install.txt 2>&1
```

Again, this process may take 30-45 minutes to run, depending on your hardware and the HEASoft download selections you choose. If it is successful, the last line of output (in install.txt) should read "Finished make install". Check it using the command:

```
less install.txt
```

5.1.6 Initialize

Once the installation is successful, all that remains is to initialize it. For this, you will need to specify the path you are working on. The PLATFORM indicates your machine's architecture. You will find it inside the folder **heasoft6.36** with a name like **x86_64-pc-linux-gnu-libc2.39**. Copy its path and place it below.

```
export HEADAS=/path/to/your/installed/heasoft-6.36/(PLATFORM)
source $HEADAS/headas-init.sh
```

To make sure that the HEADAS path is exported every time you open the terminal, paste the export line inside your source file. Here's how to do it.

```
vi ~/.bashrc
```

Paste the line below at the last of the file.

⁶If it is not successful, do not proceed until any errors have been resolved.

```
export HEADAS=/path/to/your/installed/heasoft-6.36/x86\_64-pc-linux-gnu-libc2.39
```

And save it. Make sure you replace the actual path and platform name. You can check your path by `pwd` command.

5.1.7 Testing

The initialization is silent and only generates a message if there is an error. If it is successful, you may begin using the software. To check if it is installed, type this in the terminal

```
xspec
```

If it is successful, it should show something like this:

```
XSPEC version: 12.15.1
Build Date/Time: Thu Jan 29 09:46:31 2026

XSPEC12>
```

And you are done installing HEASoft. Congratulations on completing this step.

5.2 Installing CALDB

The Calibration Database (CALDB) is required for data reduction and is integrated with the HEASoft software package in that the HEASoft tasks can access appropriate calibration data from the CALDB with little input from the user. Installing all or part of the CALDB is a matter of setting up the CALDB environment variables and downloading and untarring the calibration data. These steps are described below.

5.2.1 Setting Up Environment

Define the environment variable `$CALDB` to point to the directory where the calibration data will be located. For this create a new directory inside your home with name `caldb`.

```
mkdir caldb
```

This directory is also called the top directory. Then set up the environment in that directory.

```
export CALDB=/home/subhash/caldb
```

Make sure you replace *subhash* with your device username.

5.2.2 CALDB Setup Files

Now, we will change to the CALDB directory which we just created in the above section, and download the setup files. Run the following commands to do so.

```
cd $CALDB
wget https://heasarc.gsfc.nasa.gov/FTP/caldb/software/tools/ \
    caldb_setup_files.tar.Z
```

Once, the download is complete, extract the tar file.

```
gunzip caldb_setup_files.tar.Z
tar -xvf caldb_setup_files.tar
```

This will create the following files with the structure:

```
$CALDB/software/tools/alias_config.fits
$CALDB/software/tools/caldb.config
```

along with the initialization files

```
$CALDB/software/tools/caldbinit.csh
$CALDB/software/tools/caldbinit.sh
```

Now, open the appropriate initialization file `caldbinit.sh`, in an editor (like `vi` or a simple text editor), and edit the definition of the `$CALDB` environment variable to point to your top-level CALDB directory. To do this, uncomment and edit lines 12, 13, and 14 of the file `caldbinit.sh` as shown below:

```
CALDB=/home/subhash/caldb; export CALDB
CALDBCONFIG=$CALDB/software/tools/caldb.config; export CALDBCONFIG
CALDBALIAS=$CALDB/software/tools/alias_config.fits; export CALDBALIAS
```

5.2.3 Setup

Before accessing calibration data, you need to set up the CALDB environment variables by sourcing the appropriate CALDB initialization file. Run this command to source the initialization file.

```
source $CALDB/software/tools/caldbinit.sh
```

5.2.4 Installing NuSTAR Calibration Data

Now we will learn to download calibration data from the HEASARC for a particular mission, NuSTAR in our case, and instrument, and how to store the data so the CALDB software tools can access it. Follow these steps to do so:

1. Go to the top-directory CALDB which we have created in the previous section.
2. Visit the website https://heasarc.gsfc.nasa.gov/docs/heasarc/caldb/caldb_supported_missions.html and find the table Supported Missions/Instruments
3. Scroll until you find the NuSTAR row, and click on the tar file on 5th column.
4. The downloading of tar file named `goodfiles_nustar_fpm_20251215.tar.gz` will start.
5. Once finished downloading, untar it

```
tar -zxvf goodfiles_nustar_fpm_20251215.tar.gz
```

This will create the CALDB directory structure for the NuSTAR mission and its instrument with the currently-available calibration data.

6. Now we can run the FTOOL `caldbininfo` to check that our installation for this mission and instrument is correct. Use the command-line syntax

```
caldbininfo INST NuSTAR FPM
```

If completed, it will show the version of CALDB and the supported mission.

5.3 Installing DS9

DS9 is a visualization software that we will use to visualize event files and extract source and background regions from them. Installing DS9 is simple and can be done in multiple ways.

1. Go to official website of SAOImageDS9: <https://sites.google.com/cfa.harvard.edu/saoimageds9>
2. From the navigation menu, click on Download.
3. Select your operating system and click on the required distribution. For me it's Linux > Ubuntu > Ubuntu 22 ARM64
4. Once the download is complete, unpack or unzip it, and you are ready to go.

Alternatively, in Linux, you can directly install via your command prompt

```
sudo apt-get install ds9
```

To check whether your installation is complete, type `ds9` in your terminal. It will open a user interface of `ds9`. If it fails, it will return an error message. Now, it's your duty to find out why it failed. Not my fault, though.

Suggestion

It is recommended to use DS9 but there are many softwares from which you can visualize and extract the information of the source. You are free to use any of them.

6 Data Cleaning and Calibration

We have successfully entered into the data analysis part now. By this time, let us recheck what we have already done:

- downloaded and successfully installed the latest version of the HEASoft X-ray data analysis software package
- successfully set up the NuSTAR calibration database (CALDB)
- downloaded the NuSTAR observation(s) to be processed from the NuSTAR archive at HEASARC.

- successfully installed the SAOImage DS9 astronomical imaging and data visualization application that can display FITS format images.

If you have passed the checkpoints, you are ready to go. If not, please complete those steps and come back again. As mentioned earlier, the data analysis steps contain three stages: data calibration, data screening, and product extraction. We will use the shortcut method by using the `nupipeline` command to do the first two stages and the `nuproducts` command to extract the scientific products.

6.0.1 nupipeline

Also, remember, while downloading data for the source, we have taken MAXI J1820+070 as an example. So, I will assume you have an observation id 90401309039 ready on your file directory. Then you are all ready to go. As mentioned earlier, we will use `nupipeline` for standard pipeline processing, and to do so, all that is needed is to specify the input path to the top-level directory (90401309039 in our case) containing the observation to be processed, the stem name of the data files, and an output directory. Run `nupipeline` stages 1 and 2 by issuing the command:

```
nupipeline indir=/90401309039/ steminputs=nu90401309039 outdir=./out
```

Now, go to the output directory that was produced

```
cd ./out
```

The level 2 output files are the calibrated, cleaned, and screened events files from the NuSTAR X-ray telescope. The following files in the output directory contain the science quality X-ray events. Make sure you have these files in `out` directory.

```
nu90401309039A01_cl.evt  
nu90401309039B01_cl.evt
```

where A and B in the above event files indicate FPMA and FPMB science quality event FITS file. The `nupipeline` script can be used to automatically run stage 3 to extract level 3 products, but the default use is to run stages 1 & 2. We will run stage 3 separately via the `nuproducts` script in the next section.

6.0.2 ds9

Now, we will extract the source and background regions from the obtained event files. First, go to the output directory,

```
cd ./out
```

Open event file `ds9`,

```
ds9 nu90401309039A01_cl.evt
```

Make source extraction region files using DS9 by choosing a circular region of 30 arcsec radius centered on the source, similar to what is shown in Figure 3. Save it as `src.reg`. Choose the WCS coordinate system when saving the region files. Follow these steps to do so,

1. Go to Region Tab.

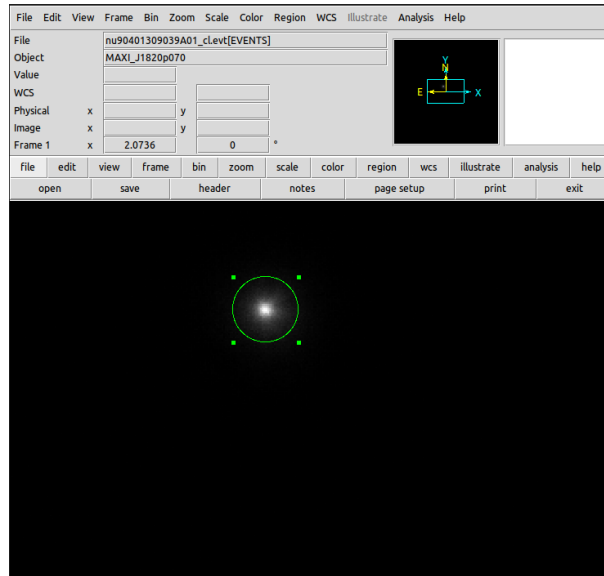


Figure 3: Source region with 30 arcsec radius centered at source peak brightness

2. Click on Shape and select Circle.
3. Double click on the source's centre and drag outwards until it encloses the source completely.
4. Now, single-click on the circle and go to the region tab.
5. Click on get information and enter 30 on radius and make ArcSec and click apply.
6. Close the tab and again go to the region and click on save selection. Rename the file as **src.reg**. In the Coordinate System, change to WCS.

To check if your extraction is complete, open the region file and make sure you have a similar line:

```
circle(275.0939417,7.1849453,47.985")
```

Similarly, extract the background region and save it as **bkg.reg**.

6.0.3 nuproducts

Once finish extracting both source and background files, we are ready to run products extracting commands. We will use **nuproducts** command to do so.

```
nuproducts \
  srcregionfile=src.reg \
  bkgregionfile=bkg.reg \
  indir=./out \
  outdir=./products \
  instrument=FPMA \
  steminputs=nu90401309039 \
  bkgextract=yes
```

You will get the scientific product files as summarized in Table 2. Once done, we are ready to analyze the files using XSPEC.

File Type	What do they mean?
nu90401309039A01_sk.img	FITS Science mode image
nu90401309039A01_im.gif	GIF Science mode gif image
nu90401309039A01_sr.pha	FITS Source energy spectrum
nu90401309039A01_ph.gif	GIF Source energy spectrum plot
nu90401309039A01_sr.lc	FITS Source light-curve
nu90401309039A01_lc.gif	GIF Source light-curve plot
nu90401309039A01_bk.pha	FITS Background energy spectrum
nu90401309039A01_bk.lc	FITS Background light-curve
nu90401309039A01_sr.arf	FITS Source ARF file
nu90401309039A01_sr.rmf	FITS Source RMF file

Table 2: extracted scientific products files

7 Analysis using XSPEC

Once X-ray data are extracted and response matrices are produced, we will prioritize the following files during XSPEC analysis,

```
nu90401309039A01_sr.arf      nu90401309039_sr.pha
nu90401309039A01_sr.rmf
```

7.1 Loading Data

Typing `xspec` will open the XSPEC command line interface. Now enter the following commands to load the data.

```
XSPEC12>data nu90401309039_sr.pha
```

This command will automatically use response files and auxiliary response files from your directory. Now, run the following commands to get a basic overview and plot of the data.

```
XSPEC12>ignore bad
XSPEC12>ignore **-3.0 79.-**
XSPEC12>cpd /xw
XSPEC12>plot ldata
XSPEC12>setplot command r y
```

You can now view the summary of the data by following the command.

```
XSPEC12>show all
```

7.2 Fitting a Model

Before applying any model, you can find the value of the hydrogen column density `nh`, which quantifies the amount of interstellar absorption affecting the X-ray spectrum, by entering the following command.

```
XSPEC12>nh
```

Now, let's apply a simple model which combines absorption and powerlaw model. You can find a list of models in the XSPEC page <https://heasarc.gsfc.nasa.gov/docs/software/xspec/manual/Models.html>. Let's fit a simple model. Here's how you do it.

```
XSPEC12>mo pha*po
```

Input the required values for each parameter and proceed further.

```
XSPEC12>query yes  
XSPEC12>renorm  
XSPEC12>fit 100
```

View the fitted spectra,

```
plot ldata delchi
```

You can add other components to your model, too. To do this, simply run,

```
XSPEC12>addcomp 2 powerlaw  
XSPEC12>fit 100
```

The above commands add two components of the power law to account for the residuals observed at low energies and fit the spectra again.

More on XSPEC Analysis

Due to the time limit, I have not included everything about XSPEC, its analysis, and interpretation. You can find the guide presentation for some reference. Visit the workshop site sxpwoac2026.github.io/notes for more info.

8 Important Sources

This manual is straightforward and skips many steps, which are often not that important for analysis. But for advanced analysis and other missions, you may want to look up references that are important to understand the workflow. So, I have collected some useful references you may want to look up to further explore this field. Clicking on them will automatically navigate you to the source.

- [HEASoft Installation Guide - Linux](#)
- [HEASoft Installation Guide - MacOS](#)
- [XSPEC Users Guide](#)
- [CALDB Installation Instructions](#)
- [CALDB Supported Missions](#)
- [SAOImageDS9 Users Manual](#)
- [Models in XSPEC](#)

9 Conclusion

In conclusion, this Workshop on Astrophysics and Cosmology, organized by SXPC-NEPAL, has provided participants with a comprehensive introduction to two major fields of modern astronomy. Through hands-on sessions, we delved into the physics of black hole low-mass X-ray binaries (BH LMXBs) by analyzing real NuSTAR observations with HEASOFT, NuSTARDAS, XSPEC, and CALDB, and investigated the large-scale structure and expansion history of the Universe through cosmological data fitting and contour analysis of Hubble datasets. These practical exercises not only built technical skills in data handling, spectral modeling, and statistical inference, but also illustrated how observations across vastly different scales, from nearby planetary systems to distant supernovae and accreting black holes, contribute to our understanding of the cosmos. We hope that the knowledge, tools, and curiosity gained during these three days will inspire you to continue exploring these fascinating topics, whether through further self-study, participation in research projects, or future advanced workshops. Thank you all for your active engagement. The universe still holds many secrets, and we look forward to seeing where your astronomical journey takes you next.

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